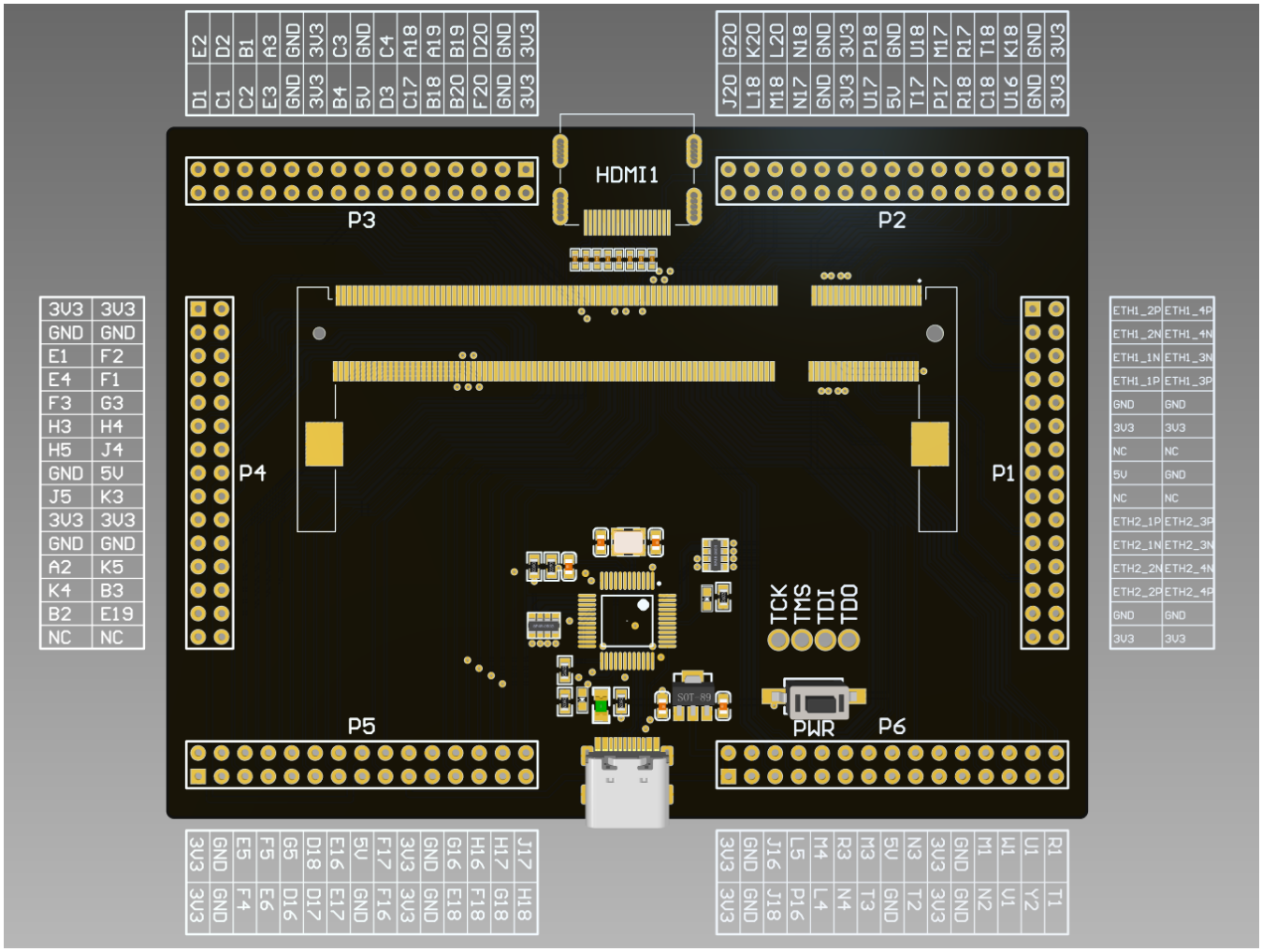


Component

- FPGA
  LFE5U-25F-6BG381C
- SDRAM
  EM638325BK-6H 8MB
- SPI Flash
  GD25Q16CSIG 2MB
- Ethernet PHY
  1Gb Ethernet PHY Broadcom B50612D x 2

JTAG

Pin	Function
J27	TCK
J31	TMS
J32	TDI
J30	TDO

Clock

A 25Mhz clock is connected to FPGA with pin P3

LED

Pin	Function
U16	D2

SPI-FLASH (U31)

Pin	Function
R2	CS
V2	MISO
W2	MOSI
U3	SCK

DDR2-SODIMM-200P

Function	Top-Pin	Bot-Pin	Function
GND	1	2	5V
GND	3	4	5V
GND	5	6	5V
GND	7	8	5V
GND	9	10	5V
GND	11	12	5V
NC	13	14	NC
ETH1_1P	15	16	ETH2_1P
ETH1_1N	17	18	ETH2_1N
NC	19	20	NC
ETH1_2N	21	22	ETH2_2N
ETH1_2P	23	24	ETH2_2P
NC	25	26	NC
ETH1_3P	27	28	ETH2_3P
ETH1_3N	29	30	ETH2_3N
NC	31	32	NC
ETH1_4N	33	34	ETH2_4N
ETH1_4P	35	36	ETH2_4P
NC	37	38	NC
GND	39	40	GND

Function	Top-Pin	Bot-Pin	Function
U16	41	42	R1
NC	43	44	T1
NC	45	46	U1
NC	47	48	Y2
K18	49	50	W1
C18	51	52	V1
NC	53	54	M1
GND	55	56	GND
T18	57	58	N2
R18	59	60	N3
R17	61	62	T2
P17	63	64	M3
M17	65	66	T3
T17	67	68	R3
U18	69	70	N4
U17	71	72	M4
P18	73	74	L4
N17	75	76	L5
N18	77	78	P16
M18	79	80	J16
L20	81	82	J18
L18	83	84	J17
K20	85	86	H18
K19	87	88	H17
J20	89	90	G18
J19	91	92	H16
H20	93	94	F18
G20	95	96	G16
G19	97	98	E18
F20	99	100	F17
F19	101	102	F16
E20	103	104	E16
GND	105	106	GND

Function	Top-Pin	Bot-Pin	Function
GND	107	108	GND
E19	109	110	E17
D20	111	112	D18
D19	113	114	D17
C20	115	116	G5
B20	117	118	D16
B19	119	120	F5
B18	121	122	E6
A19	123	124	E5
C17	125	126	F4
A18	127	128	E4
D3	129	130	F1
C4	131	132	F3
B4	133	134	G3
C3	135	136	H3
E3	137	138	H4
A3	139	140	H5
C2	141	142	J4
B1	143	144	J5
C1	145	146	K3
D2	147	148	K4
D1	149	150	K5
E2	151	152	B3
E1	153	154	A2
F2	155	156	B2
GND	157	158	GND
NC	159	160	NC
NC	161	162	NC
NC	163	164	NC
NC	165	166	NC
NC	167	168	NC
NC	169	170	NC
NC	171	172	NC

Function	Top-Pin	Bot-Pin	Function
NC	173	174	NC
NC	175	176	NC
NC	177	178	NC
NC	179	180	NC
NC	181	182	NC
NC	183	184	NC
NC	185	186	NC
NC	187	188	NC
NC	189	190	NC
NC	191	192	NC
NC	193	194	NC
NC	195	196	NC
NC	197	198	NC
GND	199	200	GND

SDRAM (U18)

Pin	Function
B9	CLK
VCC	CKE
GND	CS
B10	RAS
A9	CAS
A10	WE
GND	DQM0
GND	DQM1
GND	DQM2
GND	DQM3
B11	BA0
(GND on v6.0; C8 on v7.0)	BA1
B13	A0
C14	A1
A16	A2
A17	A3

Pin	Function
B16	A4
B15	A5
A14	A6
A13	A7
A12	A8
A11	A9
B12	A10
B6	DQ0
A5	DQ1
A6	DQ2
A7	DQ3
C7	DQ4
B8	DQ5
B5	DQ6
A8	DQ7
D8	DQ8
D7	DQ9
E8	DQ10
D6	DQ11
C6	DQ12
D5	DQ13
E7	DQ14
C5	DQ15
C10	DQ16
D9	DQ17
E11	DQ18
D11	DQ19
C11	DQ20
D12	DQ21
E9	DQ22
C12	DQ23
E14	DQ24
C15	DQ25

Pin	Function
E13	DQ26
D15	DQ27
E12	DQ28
B17	DQ29
D14	DQ30
D13	DQ31

ETH-PHY0 (U28)

Pin	Function
N5	MDC
P5	MDIO
P4	RESET
U19	GTXCLK
U20	TXD[0]
T19	TXD[1]
T20	TXD[2]
R20	TXD[3]
P19	TX_EN
L19	RXC
P20	RXD[0]
N19	RXD[1]
N20	RXD[2]
M19	RXD[3]
M20	RX_DV

ETH-PHY1 (U29)

Pin	Function
N5	MDC
P5	MDIO
P4	RESET
G1	GTXCLK
G2	TXD[0]
H1	TXD[1]

Pin	Function
J1	TXD[2]
J3	TXD[3]
K1	TX_EN
H2	RXC
K2	RXD[0]
L1	RXD[1]
N1	RXD[2]
P1	RXD[3]
P2	RX_DV

How to Buy

you can buy Colorlight i5 and 5A-75B on our aliexpress store
Colorlight i5 & i9
Colorlight i9plus Colorlight 5A-75B

reference

- chubby75
  https://github.com/q3k/chubby75
- Colorlight-5A-75B
  https://github.com/kholia/Colorlight-5A-75B
- colorlight-i5-tips by kazkojima
  https://github.com/kazkojima/colorlight-i5-tips
- ulx3s_examples by lawrie
  https://github.com/lawrie/ulx3s_examples
- ULX3S-Blinky by DoctorWkt
  https://github.com/DoctorWkt/ULX3S-Blinky
- litex without dram
  https://github.com/wuxx/litexOnColorlightLab004
- litex with dram
  https://github.com/ghent360/riscvOnColorlight-5A-75B
- litex with ethernet
  https://github.com/enjoy-digital/colorlite
- supported in IceStudio and Apio https://github.com/benitoss/ColorLight_FPGA_boards